CSET485 AI and Society, Assignment 1

Stress-testing an income classifier on the Adult Census data

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Dataset	Adult Census Income, UCI (roll number ends in 8)
Random seed	138 (last three digits of roll number), 70% train / 30% test
Code	https://github.com/sanskar-94/income-classifier-stress-test

Every number in this report comes from the notebook `income_classifier_stress_test.ipynb` in the repository above. Running it from top to bottom recreates all tables and figures.

Data preprocessing

The Adult data (1994 US Current Population Survey) comes as `adult.data` (32,561 rows) and `adult.test` (16,281 rows). I stacked both files (48,842 rows), removed the trailing full stop from the test labels (">50K."), dropped 52 exact duplicates (48,790 rows left) and made my own 70/30 split with `random_state = 138`: 34,153 training rows and 14,637 test rows, with 23.9% and 24.0% earning >50K.

Missing values. "?" appears in workclass (2,799), occupation (2,809) and native country (857); 3,620 rows (7.4%) have at least one. I kept these rows because missingness is linked to income: 13.2% of them earn >50K against 24.8% of complete rows. Each missing value gets the training-set mode, plus a 0/1 flag recording that it was missing. Workclass and occupation are missing on the same rows, so they share one flag (`job_missing`); native country has `country_missing`. A first attempt with separate "Unknown" levels for both columns did not converge, because the two dummies differ on only 9 training rows (coefficients +14.2 and -15.5, standard errors about 5,200).

Columns left out. `fnlwgt` is a survey weight, not a property of the person. `education` duplicates `education_num`. `relationship` is mostly marital status crossed with sex ("Husband" is 19,715 men and 1 woman), so it would blur the sex coefficient; I only use it for a residual check in Question 4.

Categorical encoding. One-hot encoding with one baseline level per variable (White, Female, Never-married, Private, Prof-specialty, US). Small categories were merged first: the three government workclasses into "Government", "Never-worked" and "Without-pay" into "No-pay", "Armed-Forces" into "Protective-serv", marital status into four levels (Married, Never-married, Div-Sep, Widowed) and 41 countries into US / Non-US. The model has 32 predictors plus an intercept.

Outliers. Nothing was deleted. The 1.5 × IQR rule flags 13,486 hours values (27.6%, everyone outside 32.5 to 52.5 hours), 215 ages (79 to 90) and 1,787 low education levels, all plausible. For capital gain and loss the IQR is zero (91.7% and 95.3% zeros), so the rule is meaningless there. Age 90, 99 hours and a $99,999 capital gain are survey top codes. The real issue is the long tail of the capital columns, so I applied `log1p` to both.

Feature scaling. The five numeric columns are standardised with a scaler fitted on the training rows only (imputation modes also come from training rows), so nothing leaks from the test set. Dummies stay 0/1, so numeric coefficients are per one standard deviation (13.7 years of age, 2.6 education levels).

Part 1: Build

Model A is OLS on the 0/1 income label (a linear probability model) and Model B is an unregularised logistic regression, both on the same 32 predictors and training rows. They rank test-set people almost identically (Spearman correlation 0.984; AUC 0.894 and 0.898), but 21.6% of Model A's predictions are below zero, so they are not real probabilities.

The median of Model A's test predictions is **0.2262**; Model A flags everyone at or above it (7,319 people, 50% by construction). Model B flags $p \geq 0.5$ (2,848 people, 19.5%).

Table 1. Agreement between the median rule (Model A) and the 0.5 rule (Model B) on the 14,637 test rows.

	B low (p < 0.5)	B high (p >= 0.5)	Total
A low (< median)	7,318	0	7,318
A high (>= median)	4,471	2,848	7,319
Total	11,789	2,848	14,637

The rules disagree on **4,471 people (30.55%)**, all of them "high" under Model A and "low" under Model B. Every person Model B flags is also flagged by Model A, so the mismatch is simply 7,319 minus 2,848: the median rule sets a much lower bar than 0.5 and flags half the test set when only 24.0% earn >50K. Its accuracy is 71.3%, against 84.6% for the 0.5 rule (see Question 1).

Part 2: The Trap

2.1 Simpson's paradox hunt

I split the training data by marital status (Subgroup A: married, 15,735; Subgroup B: unmarried, 18,418) and track the coefficient on `sex_Male` (baseline Female), fitting the full Model B specification on all training rows and on each subgroup. In the unmarried subgroup two dummies with no high earners at all (private household service, no-pay) were dropped so the fit converges.

Table 2. Share earning >50K in the training set, by marital status and sex.

	Women	Men
Married (Subgroup A)	44.7% (n = 1,746)	44.3% (n = 13,989)
Not married (Subgroup B)	4.7% (n = 9,494)	8.3% (n = 8,924)
All training rows	11.0% (n = 11,240)	30.3% (n = 22,913)

Table 3. Logit coefficient on sex_Male, with 95% confidence intervals.

Model	n	Crude coef. [95% CI]	Adjusted coef. [95% CI]	p (adjusted)	Odds ratio (adjusted)
Full training set	34,153	+1.262 [1.196, 1.328]	**+0.160** [0.064, 0.256]	0.001	1.17
Subgroup A: married	15,735	-0.017 [-0.117, 0.083]	**-0.288** [-0.416, -0.161]	< 0.001	0.75
Subgroup B: unmarried	18,418	+0.599 [0.478, 0.720]	**+0.740** [0.584, 0.895]	< 0.001	2.10

Result. The adjusted coefficient is +0.160 overall, reverses to **-0.288 (CI -0.416 to -0.161)** among married people and rises to +0.740 among unmarried people. The subgroup difference is far too large to be chance (z = -10.03; interaction term -1.191, CI -1.376 to -1.006). Refitting on all 48,790 rows gives the same pattern (+0.151, -0.287, +0.689).

Explanation. The crude gap is mostly composition: 61.1% of men but only 15.5% of women are married, and married people earn >50K far more often. The adjusted reversal happens because Model B fits one sex effect for everyone, so +0.160 is a compromise between opposite effects in the two groups (Question 2).

Policy implication. Ignoring the subgroups suggests a small, uniform male advantage (odds ratio 1.17). The subgroups show it is concentrated among unmarried people (odds ratio 2.10), so measures on the pay gap should target unmarried women first.

2.2 Omitted-variable bias

I dropped marital status, the strongest predictor in Model B (+2.586 for married) and one closely tied to sex, age and race.

Table 4. Coefficients before and after removing marital status. % change is (reduced - full) / |full|.

Predictor	Full model	Without marital status	Change	% change	Odds ratio
sex_Male	0.160	1.132	+0.972	+607.9%	1.17 → 3.10
age	0.362	0.583	+0.221	+61.2%	1.44 → 1.79
race_Black	-0.237	-0.459	-0.222	-93.7%	0.79 → 0.63
workclass_Self-emp-inc	0.179	0.312	+0.133	+74.0%	1.20 → 1.37
occupation_Exec-managerial	0.212	0.270	+0.059	+27.7%	1.24 → 1.31
hours_per_week	0.368	0.390	+0.021	+5.8%	1.45 → 1.48
education_num	0.757	0.682	-0.075	-10.0%	2.13 → 1.98
capital_gain	0.514	0.498	-0.016	-3.1%	1.67 → 1.65

Which shifted most. `sex_Male`: 0.160 to 1.132 (+607.9%), also the largest absolute change (+0.972) of the 29 shared coefficients. `race_Black`, incorporated self-employment and age followed (-93.7%, +74.0%, +61.2%), each in the direction of its link with marriage. Education and capital gain shrank slightly, a known property of logistic regression when a strong predictor is left out.

What it reveals. The reduced model still looks fine (AUC 0.852 vs 0.898, accuracy 82.3% vs 84.6%), yet its sex coefficient is about seven times larger. Performance figures alone would never show this.

Reflection. I would look for hidden confounds with a residual audit on variables the model does not use. With the dropped `relationship` column, wives are under-predicted by 0.084 on average ($t = 5.12$), a gap that a sex × married interaction removes (Question 4).

2.3 Adversarial subset construction

Model B makes 215 confident false positives ($p \geq 0.75$, actually ≤50K) and 623 confident false negatives ($p \leq 0.25$, actually >50K): **838 confident mistakes, 5.73% of the test set**. A random sample of 50 (seed 138) has 37 false negatives and 13 false positives.

Table 5. The 50-row sample compared with all 838 confident errors and the full test set.

	50-row sample	all 838 confident errors	full test set
n	50	838	14,637
mean age	41.5	43.4	38.8
median age	38.5	41.0	37.0
age under 30 (%)	8.0	11.5	29.5
age 30-49 (%)	64.0	58.6	48.1
age 50+ (%)	28.0	30.0	22.4
male (%)	78.0	75.2	66.3
married (%)	68.0	63.7	45.6
White (%)	90.0	87.5	86.0
Black (%)	4.0	6.9	9.4
born outside US (%)	6.0	9.1	8.4
bachelor's or higher (%)	46.0	35.8	25.2
works 40+ hours (%)	90.0	86.0	75.7
any capital gain (%)	10.0	13.2	8.4
actually earns >50K (%)	74.0	74.3	24.0

Common characteristic. In 49 of the 50 rows (98%) the marriage-and-degree profile points the wrong way: 12 of the 13 false positives are married graduates (only 5.3% of test-set people earning ≤50K are), and none of the 37 false negatives is. The sample is also older, more male and more often married than the test set.

Why the model fails. The model is additive and the married coefficient (+2.586) dominates; going from high school to a bachelor's degree adds only about 1.18. Married graduates start near the 0.75 line, so a managerial job or long hours pushes them over it even when they earn ≤50K, while unmarried people start 2.1 to 2.6 behind. Adding a sex × married interaction and an age² term only cuts the confident errors from 838 to 819 (Question 5).

2.4 Fairness trade-off analysis

Setup. Sensitive variable: sex. Group A = Female (4,936 test rows, 10.9% earn >50K), Group B = Male (9,701 rows, 30.6%). With women as Group A, both EO ratios are below 1, so the minimum in the EO formula picks the larger gap. I assume the model pre-approves people for a credit product needing an income above $50K, so $\hat{Y} = 1$ is the favourable outcome. DP, EO and PP are female/male ratios (ideal 1.0); TPR, FPR and PPV below are per group.

Table 6. Fairness metrics across thresholds (F = female, M = male).

t	DP	EO	PP	Accuracy	TPR F	TPR M	FPR F	FPR M	PPV F	PPV M
0.05	0.600	0.612	0.578	0.614	0.961	0.986	0.363	0.593	0.245	0.424
0.15	0.358	0.320	0.857	0.758	0.809	0.937	0.128	0.401	0.435	0.508
0.25	0.285	0.244	0.929	0.811	0.638	0.856	0.067	0.275	0.538	0.579
0.35	0.260	0.210	0.978	0.837	0.545	0.761	0.039	0.187	0.629	0.643
0.45	0.238	0.189	0.991	0.845	0.441	0.664	0.024	0.128	0.689	0.695
0.55	0.214	0.155	1.024	0.844	0.351	0.570	0.014	0.088	0.759	0.741
0.65	0.187	0.121	1.047	0.835	0.253	0.460	0.007	0.056	0.819	0.783
0.75	0.153	0.098	1.036	0.818	0.147	0.329	0.003	0.030	0.859	0.829
0.85	0.161	0.060	1.069	0.797	0.093	0.192	0.001	0.011	0.943	0.883
0.95	0.113	0.000	1.050	0.774	0.022	0.067	0.000	0.001	1.000	0.952

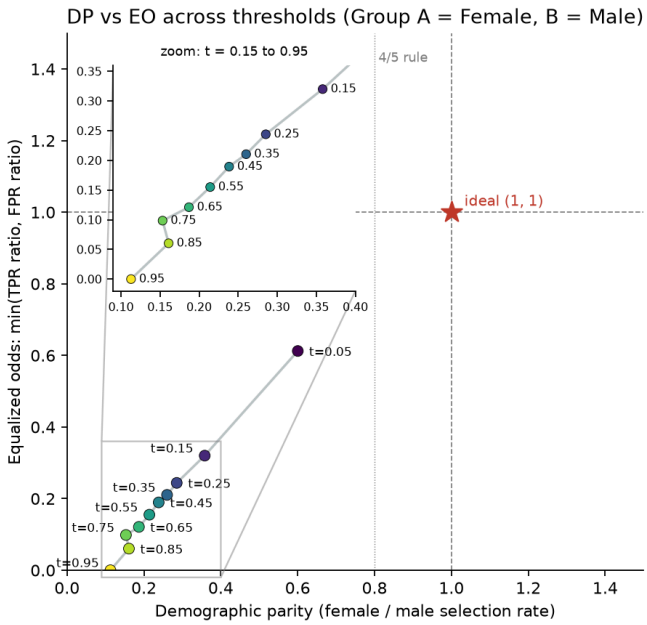


Figure 1. Demographic parity against equalized odds, one point per threshold (inset: zoom on the crowded corner). The ideal is (1, 1).

Quantifying the trade-off. DP and EO mostly rise together as the threshold falls. The one step where they conflict:

> If you improve DP from 0.153 to 0.161 (a 5.1% gain, t = 0.75 to 0.85), EO drops from 0.098 to 0.060 (a 38.7% loss). The trade-off cost is 38.7 / 5.1 ≈ 7.6.

That step rests on only 53 flagged women. The larger conflict is with predictive parity: improving DP from 0.238 to 0.358 (+50.2%, t = 0.45 to 0.15) lowers PP from 0.991 to 0.857 (-13.5%), a cost of about 0.27, and accuracy falls from 84.5% to 75.8%.

Key insight and recommendation. No threshold satisfies all three metrics: DP and EO are best at t = 0.05 (0.600, 0.612) where PP is 0.578 and accuracy 61.4%, while PP is best at t = 0.45 (0.991) where DP is only 0.238. I recommend t = 0.35 (accuracy 83.7%, PP 0.978, and better DP and EO than at 0.45), and prioritising equalized odds over demographic parity because the base rates differ so much (Questions 7 and 8).

Part 3: Descriptive analysis questions

Question 1. Why did the two approaches flag different people?

They did not flag the same people. Of the 14,637 test rows, 4,471 (30.55%) are labelled differently, and every one of them is "high" under the OLS median rule but "low" under the logistic 0.5 rule.

The cause is where each cut-off sits. The median of Model A's predictions is 0.2262, so Model A flags anyone scoring at least 0.2262, while Model B needs a probability of at least 0.5. A median rule flags half of any group by construction (7,319 people here), regardless of the base rate. Only 24.0% of the test set actually earns more than $50K, and the 0.5 rule flags 19.5% (2,848 people). All 2,848 of Model B's high-risk people are also in Model A's 7,319, so the mismatch is exactly the difference between the two counts.

The two models themselves agree closely on who is likely to be a high earner: their Spearman rank correlation is 0.984 and their AUCs are 0.894 and 0.898. The difference between the models only shows when both flag the same number of people: if Model A flagged its top 2,848, the two would disagree on just 420, because OLS and the logit weight the predictors slightly differently. Cutting the OLS scores at 0.5 instead of the median gives only 483 mismatches (3.3%).

The 4,471 people in between are uncertain cases. Their logistic probabilities run from 0.014 to just under 0.5 (median 0.247), and 28.2% of them really earn >50K, only slightly above the base rate. The median rule flags all of them, which is why its accuracy is 71.3% compared with 84.6% for the 0.5 rule.

Question 2. The subgroup with a reversed effect

I split the data by marital status and tracked the sex coefficient. In the full training set men earn >50K far more often than women (30.3% vs 11.0%); the crude coefficient on sex_Male is +1.262 (95% CI 1.196 to 1.328) and, adjusted for the rest of Model B, +0.160 (CI 0.064 to 0.256, p = 0.001).

Among married people (15,735) the sign reverses: -0.288 (CI -0.416 to -0.161), so a married woman has higher odds than a married man with the same profile (odds ratio 0.75). Among unmarried people (18,418) the coefficient is +0.740 (CI 0.584 to 0.895), much stronger than overall.

Two things explain this. The crude gap is mostly composition: 61.1% of men are married against 15.5% of women, and married people earn >50K at 44.4% compared with 6.5% for the unmarried, so much of the +1.262 is really a marriage gap. The adjusted full model already controls for marital status, so the reversal there has a different cause: Model B fits a single sex effect for everyone, and +0.160 is a compromise between the two groups. A model with a sex × married interaction shows the split directly: +0.801 for unmarried people and +0.801 - 1.191 = -0.390 for married people (interaction CI -1.376 to -1.006).

Ignoring the subgroups, a policymaker would see one small male advantage (odds ratio 1.17) and might treat the gap as minor or design one measure for all women. The subgroups point instead at unmarried women, where men's odds are about twice theirs (odds ratio 2.10). Among married people the gap runs the other way, possibly because only married women in paid work are in the data.

Question 3. The omitted-variable test

I dropped marital status (three dummies against never-married). Its married dummy has the biggest coefficient in Model B (+2.586, odds ratio 13.3), and it is linked to other predictors: 61.1% of men are married against 15.5% of women, married people are older (43.3 vs 34.6 years), and 26.9% of Black people are married against 48.1% of everyone else.

Coefficient shifts when marital status is removed (as in Table 4).

Predictor	Full model	Without marital status	Change	% change	Odds ratio
sex_Male	0.160	1.132	+0.972	+607.9%	1.17 → 3.10
age	0.362	0.583	+0.221	+61.2%	1.44 → 1.79
race_Black	-0.237	-0.459	-0.222	-93.7%	0.79 → 0.63
workclass_Self-emp-inc	0.179	0.312	+0.133	+74.0%	1.20 → 1.37
occupation_Exec-managerial	0.212	0.270	+0.059	+27.7%	1.24 → 1.31
hours_per_week	0.368	0.390	+0.021	+5.8%	1.45 → 1.48
education_num	0.757	0.682	-0.075	-10.0%	2.13 → 1.98
capital_gain	0.514	0.498	-0.016	-3.1%	1.67 → 1.65

`sex_Male` moved the most: from 0.160 to 1.132, an increase of 607.9%, taking the odds ratio for men from 1.17 to 3.10. Its absolute change (+0.972) is also the largest of the 29 shared coefficients. `race_Black` became 93.7% more negative, incorporated self-employment rose 74.0% and age 61.2%, each following its link with marriage. Education (-10.0%), hours (+5.8%) and capital gain (-3.1%) barely moved; education and capital gain even shrank slightly, because leaving a strong predictor out of a logistic regression pulls the remaining coefficients towards zero.

The reduced model is not simply wrong: its odds ratio for men (3.10) is close to the raw one (3.53), and it answers a different question, namely the sex gap when marriage is not held fixed. Strictly, marital status cannot cause someone's sex, so here it acts more like a pathway than a classic confounder. What worries me is that its performance gives no hint of this. Its test AUC is 0.852 (full model 0.898) and its accuracy 82.3% (84.6%), so nobody checking those figures would suspect that one coefficient had grown about sevenfold. I would not trust the coefficients of a model that has not been stress-tested like this, however good its accuracy looks.

Question 4. Detecting a confound I cannot see

I would use a residual audit against variables the model does not contain. If the model misses something important, people who share it will be over- or under-predicted as a group, and their average residual (actual outcome minus predicted probability) will sit away from zero.

I tested this on my own model with `relationship`, the column I dropped. On the test set, people recorded as "Wife" have an average residual of +0.084 (714 people, t = 5.12): 46.2% of them earn >50K but the model predicts 37.8%. Husbands sit at -0.003. The model under-predicts wives because it adds sex and marriage effects separately, the same problem found in Part 2.1. Adding a sex × married interaction brings their residual down to +0.006. Own-child and Other-relative still stand out after the interaction (t = -4.60 and -4.18), so some hidden structure remains.

In a real deployment I would do three things. First, ask domain experts such as a labour economist or a loan officer which income drivers the data lacks: industry, weeks worked, region, spouse's income. Second, find data that has them. The American Community Survey public-use microdata includes industry, weeks worked, region and household income, and the `folktables` package builds Adult-style income tasks from it. I would refit my model there (without the capital columns, which ACS lacks, and with an inflation-adjusted cut-off) and flag any category whose average residual is far from zero, using a Bonferroni-style threshold since many categories are tested. Third, add each flagged variable and measure how far the sex and race coefficients move, as I did with marital status.

This only catches confounders that someone has recorded, so the audit should be repeated as new data becomes available.

Question 5. The 50-row adversarial subset

The 50 rows are a random sample (seed 138) of the 838 confident mistakes in the test set (5.73%): 37 false negatives and 13 false positives.

They share a marriage-and-degree profile that points the wrong way, which is true of 49 of the 50 rows (98%). Twelve of the 13 false positives are married graduates, although only 5.3% of test-set people earning ≤50K fit that description. None of the 37 false negatives is a married graduate (this half is a weaker signal, since 59.4% of all high earners are not either).

The mechanism is the size of the marriage coefficient in an additive model. Being married adds +2.586 to the log-odds, while going from a high-school diploma to a bachelor's degree adds only about +1.18 (1.56 SD × 0.757), so the decision boundary is drawn mainly by marriage. A married graduate starts close to 0.75 (median p 0.745), and an executive or professional job or long hours pushes them over it; 12 of the 13 sampled false positives have such jobs. Married people without a degree lose much of the boost through education and occupation (machine operators get -0.889, and 10 of the 21 married false negatives are machine operators) and end up near 0.2. Unmarried people start 2.1 to 2.6 behind, so most stay under 0.25 even with a degree, which 11 of the 16 unmarried false negatives have.

Adding a sex × married interaction and an age² term only cuts the confident errors from 838 to 819. At the low end the model is calibrated on average (about 623 confident false negatives expected, 623 seen), while at the top it is overconfident (215 false positives where 169 were expected). Most of the rest probably comes from income drivers the data does not contain, such as industry or overtime.

Question 6. What the trade-off means for real people

Take two real people from the test set, in the credit pre-approval setting. The woman (39, divorced or separated, private-sector managerial job, 40 hours, $8,614 in capital gains) really earns more than $50K, but the model gives her p = 0.384. The man (38, married, same education, job type, sector and hours) earns less than $50K, yet gets p = 0.471, mostly because of the married coefficient.

At t = 0.45, the most accurate threshold in the sweep (84.5%), he is approved and she is rejected, so both decisions are wrong. At that threshold qualified women are approved 44.1% of the time against 66.4% of qualified men, and DP is 0.238.

Suppose a regulator asks for better demographic parity and the bank moves to t = 0.35. She is now approved, which is correct, and he stays approved, which is still wrong. DP rises from 0.238 to 0.260 and EO from 0.189 to 0.210; on Figure 1 this is a step up and to the right, so there is no DP-versus-EO cost here. The cost appears elsewhere: men approved without qualifying rise from 864 to 1,258, so 394 more men receive credit they may struggle to repay.

Going all the way to t = 0.05 raises DP by 151.8% (to 0.600) and lowers PP by 41.7% (to 0.578) compared with t = 0.45, the same cost of about 0.27 found in Part 2.4. Now 3,988 men and 1,596 women are approved without qualifying, and only 24.5% of approved women really earn >50K against 42.4% of approved men. A lender could use that gap to argue that approving women is riskier. Prioritising DP helped this woman, but the cost fell on the men approved without qualifying and on approved women as a group.

Question 7. Defending a threshold

I would choose t = 0.35, for four reasons.

First, the accuracy cost is small: 83.7% at 0.35, less than a point below the best in the sweep (84.5% at 0.45) and the 84.6% at the default 0.5.

Second, predictive parity is close to ideal. PP is 0.978: 62.9% of approved women really earn >50K against 64.3% of approved men, so an approval means nearly the same thing for both sexes.

Third, compared with 0.45 it helps qualified women. The share of them approved rises from 44.1% to 54.5% (TPR ratio 0.664 to 0.716), and DP and EO both improve (0.238 to 0.260 and 0.189 to 0.210). Going lower buys little: at 0.25 DP only reaches 0.285, while PP drops to 0.929 and accuracy to 81.1%.

Fourth, it suits the setting. The model flags 27.2% of the test set at 0.35, close to the 24.0% who really earn >50K.

The strongest counter-argument is that 0.35 is still very unfair on two of the three measures. A DP of 0.260 means women are approved at about a quarter of the rate of men, far below the four-fifths (0.8) guideline, and a qualified woman is still rejected 45.5% of the time against 23.9% for a qualified man. A critic could say I chose the threshold that protects the bank's accuracy and called it fair because one metric looks good. That criticism is valid: PP looks good mainly because the model is well calibrated, and a calibrated model can still treat groups very unequally. Fixing unequal error rates would need a better model or group-specific thresholds, which bring legal problems of their own.

Question 8. Demographic parity or equalized odds?

For an income model used in lending I would prioritise equalized odds, and especially its true-positive half (equal opportunity): a woman who really earns over $50K should be as likely to be approved as a man who does.

The base rates differ a lot: 10.9% of women in the test set earn >50K against 30.6% of men. In my sweep DP never goes above 0.600, and it only gets there at t = 0.05, where accuracy collapses to 61.4% and PP falls to 0.578. Even a moderate push from t = 0.45 to 0.15 raises DP by 50.2% but costs 13.5% of PP and almost nine points of accuracy (84.5% to 75.8%).

The only step where gaining DP costs EO is t = 0.75 to 0.85: DP improves from 0.153 to 0.161 (a 5.1% gain) and EO drops from 0.098 to 0.060 (a 38.7% loss), a cost of 7.6. That loss is entirely in the false-positive half; the TPR ratio actually rises from 0.446 to 0.484 and moves with DP at every step. So putting EO first mainly means judging the model on its error rates. At t = 0.35, EO is only 0.210 and the TPR ratio 0.716, which tells a policymaker that the model itself needs work, while a DP target would only adjust approval rates.

What I give up is demographic parity: women would still be approved at about a quarter of the rate of men (DP 0.260). I accept that because income does affect whether someone can repay, and approving people who cannot repay harms them as well as the lender. But the 1994 pay gap behind these base rates partly reflects discrimination, so the model should come with policy aimed at that gap.

Part 4: Reflection

"If a policy-maker asked you: 'just give me one number: should we deploy this model or not?', what would you say?"

Direct answer. I would recommend conditional deployment: the model can screen applications, but a person should make the final decision. My headline reason is that its 84.6% accuracy hides the fact that women who really earn over $50K are approved far less often than men who do.

Why a single number fails. The number a policymaker would most likely ask for is accuracy or AUC. Both look fine for this model, but neither shows the problems I found.

Overall accuracy hides who the errors fall on. At t = 0.5 the model is more accurate for women than for men, mainly because most women in the data earn under $50K and the model predicts "low" for nearly all of them. The women it gets wrong are mostly high earners it failed to approve, and in lending that is the mistake that matters most.

The threshold is not something the data can choose. My two models ranked people almost identically, yet the median rule and the 0.5 rule labelled nearly a third of the test set differently. Choosing a threshold means deciding how many wrong approvals are acceptable for each extra qualified woman approved, which is a value judgement.

A fairness score would not settle it either, because the metrics disagree. Below t = 0.45, lowering the threshold improved demographic parity and equalized odds but made predictive parity and accuracy worse, and no threshold got all three close to 1. Any combined score would depend on how the metrics were weighted.

The confident mistakes are not spread evenly. Most wrong predictions at $p \geq 0.75$ and $p \leq 0.25$ belong to people whose marital status and education point one way while their income points the other, and at the top end the model is more confident than it should be. A single accuracy figure cannot say who these people are or why the model is so sure about them.

Finally, confounds outside the data cannot be ruled out. When I removed marital status, a variable I did have, the sex coefficient grew about sevenfold while the model still looked usable. A variable that was never collected could distort the model in the same way without any metric warning me.

Grounding in evidence. At t = 0.5 accuracy is 91.8% for women and 81.0% for men, but the true-positive rates are 40.1% and 61.8%. The two thresholding rules disagreed on 4,471 people (30.55%) despite a rank correlation of 0.984. Across thresholds DP ranged from 0.113 to 0.600 and PP from 0.578 to 1.069. There were 838 confident errors (5.73% of the test set), and in 49 of the 50 sampled ones marriage and education pointed the wrong way. Dropping marital status moved sex_Male from 0.160 to 1.132 (+607.9%) while AUC only fell from 0.898 to 0.852.

Closing. I expected this analysis to end in a yes or a no. Instead, most of the answer depends on choices the data cannot make, mainly the threshold and which fairness definition to use. So rather than one number, I would give the policymaker Table 6 and explain which side of each trade-off I chose, and why.